DO I WANT TO TALK TO YOU?

Echo chambers in the subreddit r/DebateVaccines

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ABSTRACT

This work consists in the analysis of the interactions between Reddit users about a very divisive topic: vaccines. The discussions were taken from the subreddit r/DebateVaccines and analyzed by using social network science and natural language processing techniques. The goal was to understand whether we could identify echo chambers in such an environment. The study was divided in four parts: firstly, we built the network using the interactions between users in the comments section and we proceeded to study its structure. Secondly, we studied how the network could be split in different communities. Then, we checked the activity of the same users from the point of view of the submissions in the subreddit thanks to hypergraphs. We then ended with the pursuit of echo chambers. [1]

KEYWORDS

Vaccines, Debate, Reddit, Network Science, Sentiment Analysis

ACM Reference Format:
Matteo Zucchelli, Lavinia Rotellini, and Jacopo Gneri. 2023. DO I WANT TO TALK TO YOU?: Echo chambers in the subreddit r/DebateVaccines. In *Social Network Analysis '24.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

The debate on vaccines started many years before the the advent of social networks in the third millennium or the outbreak of COVID-19 in 2020: in the edition of May 1910 of the magazine called Ladies' Home Journal we can find an essay called 'The fallacy of vaccination', written by the president of the Anti-Vaccination League of America, and in its following issue of June we can read an article entitled 'What Vaccination has Really Done' by Dr. Jay Frank Schamberg, the chairman of the Vaccination Committee of Pennsylvania State Medical Society [1].

Nowadays, the same debate occurs publicly in social networks such as Twitter, Reddit, Facebook, where both factions share their ideas through posts and comments. This leads not only to the spreading of news and misinformation, but also to the creation of highly polarized communities made up of like-minded people that tend to ignore the opposite opinion [7].

With this work, we want to study how the users of online social media that have different sentiment about this topic interact: the first section is focused on how the data is collected and how the graph is built and studied; the main topic of the second part is community detection, which has been conducted using four different machine learning algorithms, while the third one concentrates on high-order interactions between users and the submissions inside the subReddit we chose for this study. The final part of this work involved NLP techniques, firstly with the aim of identifying the sentiment of each user (positive, negative or neutral about the vaccine) and then to study the activity of each kind of users and how they interact with each other.

2 DATA COLLECTION

The data used in the following analyses comes from the social network Reddit, which allows its users to create posts and comments in specific forums called 'subreddits'. The users of Reddit (also known as 'redditors') interact inside a subReddit to build deep discussions about a specific topic by sharing textual posts, images and videos.

2.1 Selected Data Sources

For our analyses, the choice has fallen on the subReddit r/DebateVaccines that contains submissions and comments regarding vaccines.

2.1.1 Crawling Methodology and Assumptions. The data has been crawled thanks to the website the-eye.eu, which stores subreddits posts and comments from June 2005 to December 2022. We extracted data from the files DebateVaccines_comments and DebateVaccines_submissions, that contain respectively the comments and the submissions of said subreddit.

The two JSON files have been turned into pandas dataframes so they could be more manageable in the data preparation and graph creation phase. The dataframe df_comments was originally made up of 753395 rows and 73 features, but based on the study *The Pushshift Reddit Dataset* by Baumgartner et al. [2] and on the type of analysis we were planning to conduct, we reduced the number of features. In particular, we only considered:

- *id*: the comment's identifier;
- *link_id*: identifier of the submission of the comment;
- *author*: identifier of the author of the comment;
- *body*: the comment's text;
- *parent_id*: identifier of the parent of the comment.

All the NaN values, deleted or removed authors, link_id or parent_id istances have been removed.

[1] **Project Repositories**
Data Collection: https://github.com/sna-unipi/2024_Zucchelli_Rotellini_Gneri/tree/main/data_collection
Analytical Tasks: https://github.com/sna-unipi/2024_Zucchelli_Rotellini_Gneri/tree/main/network_analysis
Report: https://github.com/sna-unipi/2024_Zucchelli_Rotellini_Gneri/tree/main/report

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SNA '24, 2022/23, University of Pisa, Italy
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ACM ISBN 978-x-xxxx-xxxx-x/YY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

The dataframe df_submissions originally contained 28738 rows and 123 features, but for the creation of the graph we only needed the features id and author, that indicate respectively the identifier of the submission and the username of its author.

This second dataframe allowed us to add to the df_comments dataframe the feature parent_author, that for each comment specifies the username of the author of the post or the submission parent.

In this way we have created the graph starting from the fact that each single user was a node and the interactions between users were represented by an edge (given two authors A and B, there is an edge if A comments on a post or comment created by B or viceversa).

3 NETWORK CHARACTERIZATION

The following analyses were performed on the entire set of nodes. For this part we mainly relied on the networkx library[3].

3.1 Degree distribution analysis

First of all we calculated the number of nodes (20458) and edges (196960) in the graph, then we obtained the degree of each node and the average degree of the network. The result is 19.255. We then plotted a graph to visualize the degree distribution in our network.

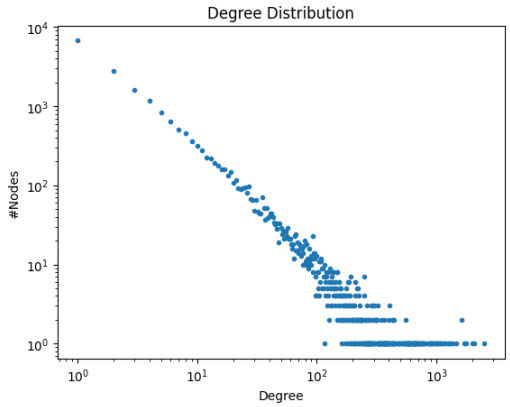


Figure 1: Degree distribution

3.2 Connected component analysis

We calculated the number of components by using the method *number_connected_components()*, and since the method found just 36 separate components we decided to print the length of the list of nodes for each one. We found out that our graph is made by a giant super component with 35 very small components in tow with a pretty consistent size between them. We plotted a graph in logaritmic scale (fig. 2) to visualize the size difference between the pieces of our network.

3.3 Global clustering coefficient and Density

The global clustering coefficient is 0.242, a low value meaning that the nodes in the graph have a low tendency to form clusters and there is an high number of weak ties between the nodes. Additionally, the density is 0.000941 suggesting that the graph is sparse and there are few connections between nodes.

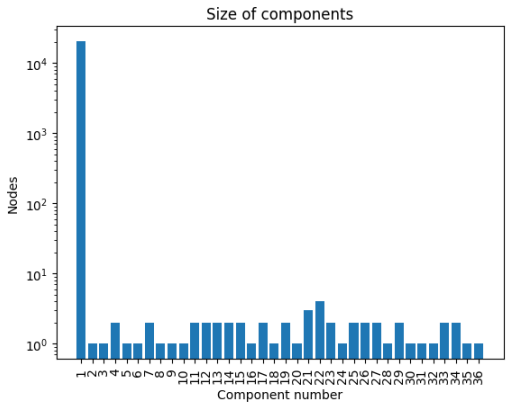


Figure 2: Size of components

3.4 Centrality analysis

After that we proceeded to compute and analyze the degree, closeness and betweenness centralities of our network. The mean degree centrality is the same value as the density (0.000941), which combined with data from the degree distribution analysis, density and clustering confirms that the graph is mostly sparse. For the closeness centrality we isolated the super component in a subgraph to let the library method do its work. This was necessary to dodge errors in the program or the code never stopping to compute. Even if this caused some error propagation in the calulations, the size of the other 35 components is so small compared to the giant one that it can be considered negligible. To have more information to reflect on we extracted some of the nodes with the highest closeness and betweenness centrality.

username	closeness cen.	betweenness cen.
jorlev	0.498	0.066
UsedConcentrate	0.484	0.058
LumpyGravy21	0.484	0.044
WWMRD2016	0.482	0.038
SftwEngr	0.482	0.036

Table 1: Top 5 users per closeness and betweenness centrality

The average closeness centrality of the graph is 0.312 while the average betweenness centrality is 0.000109: this confirms a sparse graph and, looking at the top 5 closeness and betweenness, we can see that the hubs are not so important to connect the whole network. We can imagine a graph not centralized around few nodes that play a critical role in connecting the network but, again, a sparse graph. Very interesting to note is the top 5 betweenness and closeness scores are owned by the same exact nodes.

3.5 Comparison with ER and BA graphs

To compare our graph to an analog built using ER model we calculated the probability p using the formula:

$$p = \frac{2L}{N(N-1)} = \frac{393920}{20458 \cdot 20457} = 0.00094124549765686$$

For the construction of the Barabási-Albert graph we gave the t and m values to the library function who generated the graph. The two values were calculated as shown below:

$$m = \frac{\langle k \rangle}{2} = \frac{19.255059145566527}{2} = 9.6275295727832635$$

The module did not accept a float value though, so we approximated to 10.

From the resulting inputs, the ER graph has approximately 197261 edges, while the BA graph has around 184041 edges. While that value varies for both the models one constant remains: a single connected component for both, compared to the 36 of our real graph. The degree distributions of the BA model and our real graph are pretty similar, while the ER model is completely different and represents a normal distribution (figg 4 and 3).

Comparing the results with those of the real network we noticed that the average degree in the ER and BA graph changes a little, approximately 19.331 and 19.990 compared to 19.255. The regime type of this graph is supercritical, as the value of p is greater than $1/N$, which is 0.000076. The BA graph, on the other hand, is closer to the results of the real network, with the average degree being approximately 6 (5.99) and also the degree distribution graphs are similar to those of the real network. The number of connected components is 1, resulting in a single giant component.

Tables 2 and 3 show respectively the average closeness, degree and betweenness centralities and the top 5 nodes considering closeness and betweenness for both ER and BA graphs.

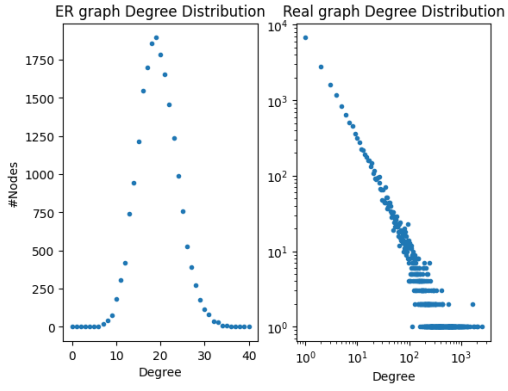


Figure 3: Degree distribution of ER graph vs Real graph

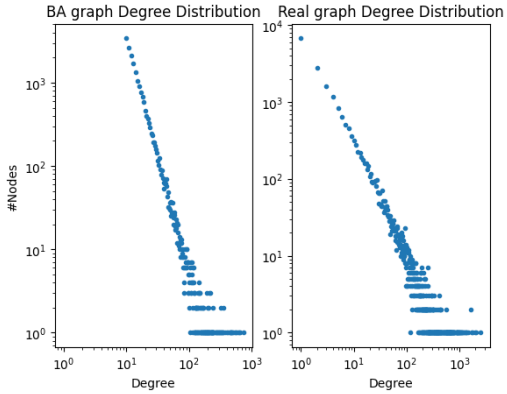


Figure 4: Degree distribution of BA graph vs Real graph

AVG	ER	BA
Degree	19.331	19.990
Closeness	0.271	0.310
Betweenness	0.000131	0.000109

Table 2: Average centralities for BA and ER graphs

ER GRAPH		BA GRAPH	
Closeness	Betweenness	Closeness	Betweenness
0.291	0.000537	0.470	0.0564
0.28856	0.000508	0.455	0.0564
0.28853	0.000495	0.443	0.0337
0.28827	0.000461	0.443	0.0329
0.28809	0.000452	0.442	0.0306

Table 3: Centrality Measures in ER and BA graphs

4 TASK 1: STATIC COMMUNITY DETECTION

In this section we aim to identify communities in our data and evaluate different methods to detect them, with the support of CDlib [6]. Four algorithms have been used to pursue this goal: Leiden (LD), Label Propagation (LP), Walktrap (WP) and Demon (D). When using these algorithms it is necessary to stress the important differences that occur between them: LD, LP and WP are all algorithms which need to be applied to an undirected graph, meanwhile Demon could be applied to a directed graph, but for semplicity reasons we decided to use an undirected one for Demon as well. Futhermore, LD, LP and WP all find crisp communities, which means that nodes cannot belong to more than one community, meanwhile the Demon algorithm finds overlapping communities, therefore a single node can belong to more than one community. Finally, the ϵ parameter (used by the Demon algorithm to determine the size of the communities) and the minimum number of communities to find were respectively set to 0.5 and 3.

4.1 Internal Evaluation

We first proceeded with the internal evaluation of the communities found by the four algorithms, which meant using the following methods: Internal Edge Density (IED), Average Node Degree (AND), Modularity (MOD) and Conductance (CON). We also share the number of communities found by each algorithm (COM) and the node coverage (NC). The results are shown in table 4 below.

Internal Evaluation Results						
Algorithm	IED	AND	MOD	CON	COM	NC
Leiden	0.33	1.25	0.34	0.20	127	1.0
Walktrap	0.05	0.1	0.25	0.96	7495	1.0
Label Propagation	0.60	1.16	0.004	0.28	316	1.0
Demon	0.06	54.83	0.21	0.21	22	0.56

Table 4: Internal Evaluation Results

Given that the Demon algorithm covers just about half of all the nodes in the community, its performance cannot really be compared to that of the other algorithms. It is also noticeable how all the algorithms partition the network in significantly different total numbers of communities. As a matter of fact, only Leiden finds communities that are proportionate, with the 98.89% of the nodes being in communities of more then 50 nodes. Yet, communities of less then 50 users are the majority even for Leiden, as it found 122 small communities versus 5 bigger ones. Label Propagation finds an even more unbalanced situation, as the 97.08% of nodes can be found in one single community, with 315 communities of less then 50 users. Walktrap finds an astonishing 7488 number of small communities, with just the 39.98% of nodes in them, it is clear how the result is very unbalanced. Demon finds mostly communities of more than 50 users, but since it covers just half the total number of nodes, it cannot be taken into consideration for further analyses.

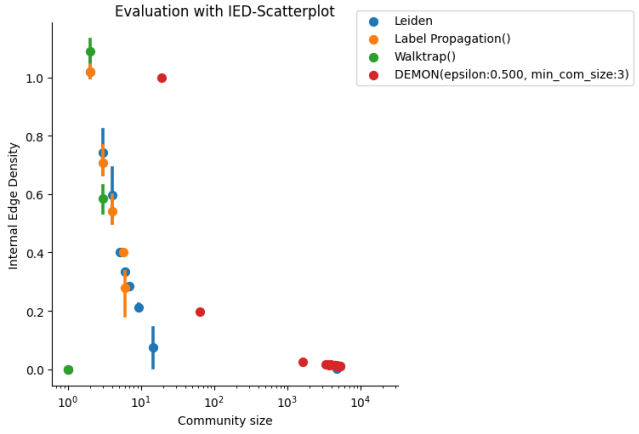


Figure 5: Internal edge density of each community found by the different partitions

Figure 5 shows how smaller communities found by Leiden, Label Propagation and Walktrap tend to have a higher internal average degree, showing how users in those communities tend to be more tightly connected. There are few exceptions to this, as seen in the communities found by the Demon algorithm and some of the communities found by Leiden.

Considering all the information we gathered, Leiden seems to be the best performing algorithm for our data as of internal evaluation, especially looking at modularity, conductance and the communities found.

4.2 External Evaluation

The external evaluation compares a partition made by an algorithm to a ground truth one, and measures how similar they are. For this evaluation we used the Adjusted Mutual Information (AMI) metric. It needs to be noted that AMI can only be computed on partitions that cover the same amount of nodes, which means that this analysis cannot be performed on the communities found by Demon. Also, we chose the partition made by Leiden as ground truth, since it was the one with the best internal statistics. Table 5 contains the results of this analysis and figure 6 shows the corrispondences between communities, as measured by AMI..

External Evaluation Results	
Partitions	AMI
LD & WP	0.19
LD & LP	0.09

Table 5: External Evaluation Results

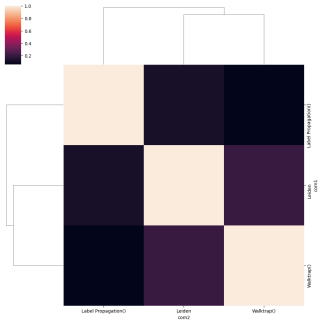


Figure 6: AMI Heatmap

Looking at AMI, it is clear how all the partitions differ between each other, and the algorithms do not find similar structures inside the data, as the correspondences are below 50%.

5 TASK 2: HIGHER-ORDER NETWORK ANALYSIS

While the previous analyses have been conducted on a simple graph where each user is represented by a node and the interaction between users is represented by an edge, the following analysis is based on hypergraphs. Their application may vary: in our case, they have been useful to investigate how users interact in online debates. In fact, for our purpose the nodes of the hypergraph are again the reddit users, while each post (or submission) is an hyperedge: an author node belongs to an hyperedge post if he commented on it. In this way, not only we can see how much each post creates a debate, but also in which way each user comments a post. We used the library HyperNetX in order to build and analyze the hypergraph, starting from a python dictionary containing for each post (as key) the set of users that commented on it (as values). The following picture shows a sample of 10 hyperedges (due to computational and visual constraints) of the hypergraph we studied, which is formed by 19856 nodes and 22609 hyperedges and by a giant connected component made of 22517 hyperedges and 85 singletons, which are posts that were commented by a single user who has commented exclusively that single post.

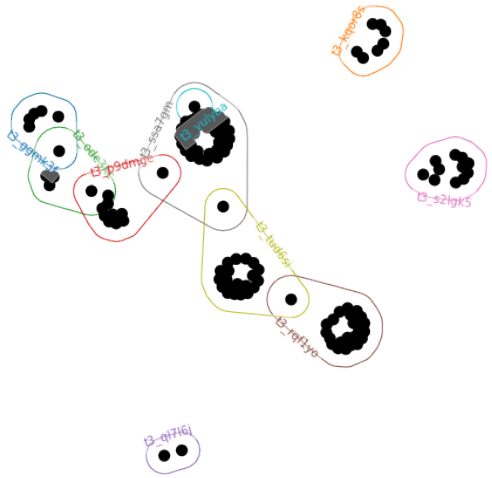


Figure 7: Hypergraph sample

In order to see how much discussion each post creates, we have studied the size of each hypergraph (figure 8).

The graph shows that the distribution is asymmetric: there is a high number of hyperedges with a small size (circa 5000 hyperedges has a size <= 2) and a small number of posts with a lot of users that comment on them: these posts are the ones that have created a lot of debate in the subreddit.

In particular, the most popular post in terms of interaction is 't3_sxgqga' created by the user oliviaisacat, that contains 178 nodes. According to the 1-linegraph of our hypergraph, the node 't3_sxgqga' has degree centrality of 10104 (the 30th in order) and a closeness centrality of 0.641: this means that it shares its nodes with almost 50% of the total posts (so, the users that commented

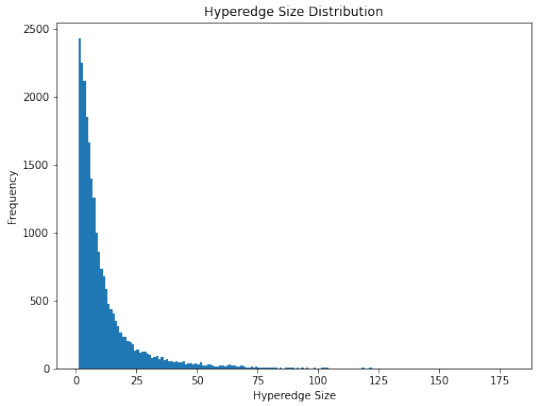


Figure 8: Hyperedge size distribution

on it create debate and comment under other posts) and that is relatively close to the other nodes in the linegraph. Moreover, 15% of the nodes of this post has degree equal to 1, which means that they have only commented on that post.

Subsequently, we analyzed the nodes degree distribution in order to see the activity, in terms of comments, of every user. Figure 9 shows that there is an high amount of nodes (9920) with degree equal to 1: those are the users that commented on a single post. On the other side, a small number of redditors comment under a high number of post: these users create more discussion compared to the previous ones.

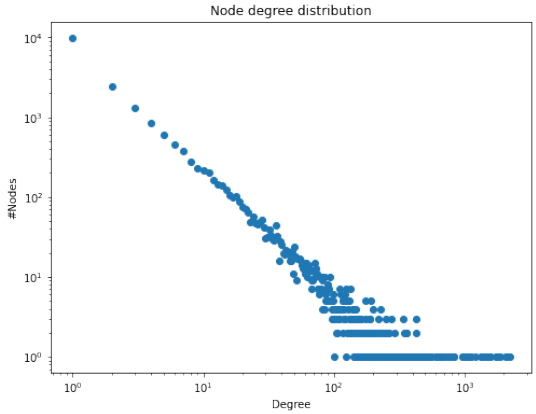


Figure 9: Node degree distribution

The most active user is 'SftwEngr', who comments under 2233 posts. Analyzing the trend of his s-neighbors (neighbors with which he shares s posts commented) we see how he has neighbors even when s > 300. In this case, is neighbor is 'WWMRD2016', with whom he comments over 300 posts.

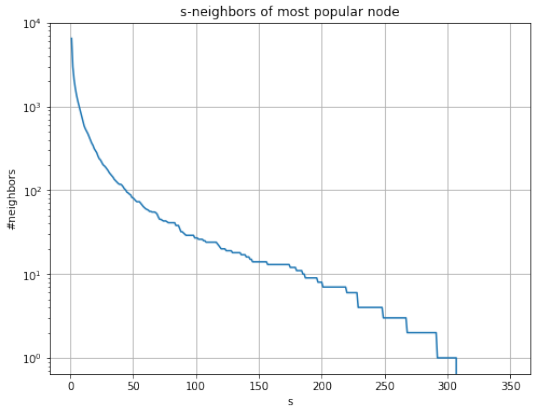


Figure 10: s-neighbors for most active user

6 OPEN QUESTION

Given the relevance of the topic and the divergence it may generate, in this final section we want to answer the following questions: how do users behave? Are they more likely to interact with people with their same feelings about the vaccines or the opposite?

In order to analyze who interacts with whom, we first had to define the sentiment of each user. Starting from the idea that users can express different feelings while posting or commenting, we used the content of their comments to decide the side of each author.

A preliminary phase of text cleaning involved the use of regular expressions to remove Urls, punctuation, double spaces and stopwords in order to prepare both datasets for the sentiment analysis.

We then used the NLTK-based Sentiment Analyzer to get the sentiments scores for "positive", "negative", "neutral" and "compound", that is a weighted sum of the previous three, with values between -1 and +1. In particular, this last score has been used in order to define the final sentiment: following [4], the sentiment is positive if the compound score is greater than or equal to 0.05, it is negative if it is lower than or equal to -0.05 and neutral otherwise.

We finally set the sentiment of each user as the most frequent sentiment in his comments and posts, resulting in 7308 positive users, 8327 negative users and 4484 neutral.

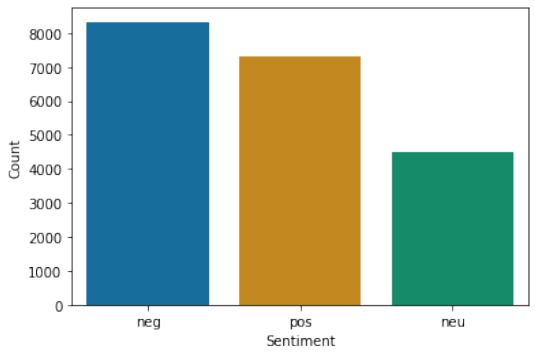


Figure 11: Users' sentiment

These labels have been attached to the nodes of the graph used in the previous analyses, and the nodes of the graph with no label have been removed.

6.1 General Measures

The first thing we checked after the graph was labeled was its homophily by using Newman's assortativity: in this case it resulted to be 0.013, which indicates that there is not assortative mixing and users do not prefer the interaction with other users they share their sentiment with.

In order to see which kind of user was the most connected, we have calculated their average degree: as table 6 shows, pos users have the highest average degree, while neg and neu have a similar score.

label	average degree
pos	44.0
neg	35.6
neu	36.1

Table 6: Label's average degree

The degree distribution does not present great differences among the labels: in every case there is a small number of nodes with a high degree and a great number of users with a low degree.

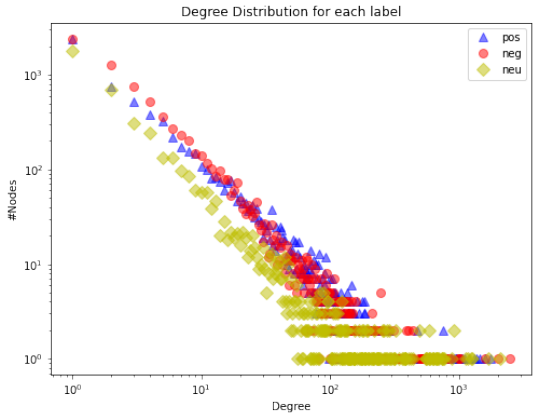


Figure 12: Degree distribution per label

We used degree centrality and closeness centrality in order to investigate on how much the users of each label interact and how close they are to each other: as table 7 shows, among the top 5 users the most active and close are neg users.

username	degree cen.	closeness cen.	label
jorlev	2495	0.497	neg
LumpyGravy21	2081	0.482	neu
UsedConcentrate	1996	0.483	neg
WWMRD2016	1766	0.480	pos
SftwEngr	1702	0.480	neu

Table 7: Top 5 users per degree and closeness centrality

6.2 Finding Echo Chambers

We then proceeded with the research of echo chambers inside our data: to do so we used the methodology proposed by Virginia Morini in her master's thesis *Polarizzazione Politica & Eco Chambers: una metodologia per l'identificazione ed analisi su Reddit* [5]. To identify echo chambers, Morini proposes to check their internal ideological and structural integrity. Since community detection was applied to the already labeled data, we decided to use **purity** as the measure for ideological integrity with a threshold of 0.7 and a *standard deviation* less or equal than 0.25. Meanwhile, to identify structural integrity, we used the EC_score, a metric proposed by Morini herself in the thesis. Below, the formula:

$$(1+\beta^2) * \frac{density_c * (1 - conductance_c)}{(\beta^2 * density_c) + (1 - conductance_c)}$$

Where $density_c$ and $conductance_c$ are respectively the internal edge density and the conductance of the community, meanwhile β is a constant used to balance the importance of the two variables in favour of conductance, and is set to 4 as per experiments made by Morini. The structural integrity is considered enough to identify an echo chamber if the EC_score is equal or above 0.5. The analysis based on this metric developed in two ways: we first considered the average EC_score for all the communities found by the algorithms, and then we computed the index just on the communities with more then 50 members, to gain more accurate insights.

6.2.1 Ideological Integrity & Purity Analysis. Purity analysis is the first step towards the identification of echo chambers, as it can help to identify specific structures and patterns. The results can be found in table 8.

Purity Evaluation Results		
Partitions	Purity	Standard Deviation
LD	0.84	0.24
WP	0.98	0.09
LP	0.80	0.24
DM	0.42	0.02

Table 8: Purity Evaluation Results

By the results it can be seen how the Walktrap algorithm has the best purity score, but since its internal evaluation values made it clear that the partitions it created are not the best ones for our data, we will proceed with the analysis of the communities found by Leiden. 92 communities were found based on purity analysis on the partition made by Leiden, meanwhile the total was of 127 communities.

	POS	NEG	NEU
# Communities	40	33	19
Avg # nodes	1.3	1.55	1.32
IED	0.15	0.24	0.12
Avg Purity	0.92	0.94	0.96

Table 9: Indices of Positive, Negative and Neutral Communities found by LD

As it can be seen in table 9, there is a majority of communities with nodes of a positive attitude, but the other values analyzed are comparable. On the other end, there is one data that is pretty interesting: the average number of nodes inside these pure communities is very low, but we will touch on this later.

6.2.2 Structural Integrity. The EC_score gave interesting information about our data, starting with the general scores as shown in table 10.

Algorithm	EC_score
Leiden	0.70
Label Propagation	0.70
Walktrap	0.04
Demon	0.36

Table 10: General EC_score Values

As it can be seen, it looks like the only algorithms that can detect structurally sound echo chambers are Leiden and Label Propagation, which were also the ones with an ideological integrity above the 0.7 threshold (Demon is not taken into consideration as it does not cover all the nodes). Since Leiden is the only one also with optimal internal evaluation statistics, we will proceed by analysing its communities. Let's look at the EC_score values Leiden gets for examples of communities that are actually relevant (i.e. the ones with more than 50 nodes) in table 11.

Algorithm	# Nodes	EC_score
Leiden	5556	0.023
Leiden	4826	0.012
Leiden	464	0.03

Table 11: Communities of >= 50 users, EC_score Values

And now let's look at examples of smaller communities of less then 50 users:

Looking at this new information we can see how bigger communities are not really echo chambers. This has a pretty simple cause: in a social network such as Reddit, communities based on interactions between users will always be mixed, as users tend to interact with the most important posts on the page, the ones that they can reach more easily, per the principle of economy. Social network analysis tells us that people tend to not search for information on

Algorithm	# Nodes	EC_score
Leiden	4	0.93
Leiden	5	0.76
Leiden	7	0.56
Leiden	2	0.70

Table 12: Communities of < 50 users, EC_score Values

social media, rather they get presented with it. On social media like Twitter and Facebook, this translates to being fed popular posts inside the social bubble a user finds themselves in, but on Reddit the most popular posts that get recommended are just the ones with the most interactions, it is not based on the preference of the user. It is the user who searches for the subReddit. Also because of the triadic closure law, smaller communities have the tendency to be more tight-knit and have a much bigger number of edges on the inside rather then towards the outside.

Our analysis therefore ends with the discovery that in our graph, built on the basis of the subReddit r/debateVaccines, echo chambers do exist, but are of a smaller dimension.

7 DISCUSSION

To conclude, this study aimed to understand whether echo chambers could be detected in a subReddit about a topic which is usually divisive. We have found that echo-chambers can in fact be found, but are not of the size we would expect. We therefore shed light on the fact that, on Reddit, users have a rather peculiar way of interacting with each other, which is different from other social media. In fact, socials such as Twitter/X or Instagram present to their users posts on their "For You" page based on their interests and history, but Reddit is primarily based on the number of interactions with a post, and it boosts the ones that are more popular. This can be seen throughout the different sections of this study, for example in the presence of one big connected component and in the small number of posts connected to a lot of users when speaking about the hypergraph. Therefore, Reddit almost fosters interactions with people of different opinions/attitudes, which means that it may not be as damaging to the public opinion as other social media.

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